Opg 2)

$0, 1, 2, \ldots, 9$

$\checkmark$

$X_1, \ldots X_{11} \sim \text{binom}(n=9;\ i=1,2,\ldots 11,\ p)$

$\hat{p}_1 = \frac{X_1}{9} \qquad \hat{p}_2 = \frac{\sum_{i=1}^{11} X_i}{9 \cdot 11} \qquad p = 0{,}6$

$$\text{Var}[\hat{p}_1] = \text{Var}\left[\frac{X_1}{9}\right] = \frac{1}{81}\text{Var}[X_1] = \frac{1}{81} \cdot np(1-p)$$

$$= \frac{1}{81} \cdot 9 \cdot 0{,}6 \cdot 0{,}4$$

$$= \frac{1}{9} \cdot 0{,}24$$

$$= 0{,}0\overline{26}$$

$$\text{Var}[\hat{p}_2] = \text{Var}\left[\frac{1}{99} \cdot \sum_{i=1}^{11} X_i\right] = \frac{1}{99^2} \sum_{i=1}^{11} \text{Var}[X_i]$$

$$= \frac{1}{99^2} \cdot 11(9 \cdot 0{,}6 \cdot 0{,}4)$$

$$= \frac{1}{99} \cdot 0{,}24 = 0{,}00\overline{24}$$

Opg 3/

$$f_x(x) = \lambda e^{-\lambda x} \quad , x>0$$

$$L(\lambda) = \prod_{i=1}^{n} f(x_i;\lambda) = \prod_{i=1}^{n} (\lambda e^{-\lambda x})$$

$$= \lambda^n \prod_{i=1}^{n} e^{-\lambda x} = \lambda^n \left(\prod_{i=1}^{n} e^{-x}\right)^{\lambda} = \lambda^n \left(e^{\sum_{i=1}^{n} x_i}\right)^{\lambda}$$

$$= \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}$$

b) $l(\lambda) = \ln(L(\lambda)) = \underline{n \ln(\lambda) - \lambda \sum_{i=1}^{n} x_i}$

c) $\frac{\partial l}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i$

d) $\frac{\partial^2 l}{\partial \lambda^2} = -n \cdot \frac{1}{\lambda^2}$

e) SME , $\frac{\partial l}{\partial \lambda} = 0 \Rightarrow \frac{n}{\lambda} = \sum_{i=1}^{n} x_i$

$$\lambda = n / \left(\sum x_i\right)$$

$$\Rightarrow \hat{\lambda} = \frac{n}{\sum_{i=1}^{n} x_i}$$

f) $\bar{x} = 8 = \frac{1}{n}\sum_{i=1}^{n} x_i$

$$\lambda = \frac{n}{\sum_i (x_i)} = 8^{-1} = 0{,}125$$

g) $E(\hat{\lambda}) = E\left(\frac{n}{\sum_i x_i}\right) = n\,E\left[\frac{1}{\sum_i x_i}\right]$

$X \sim \exp(\lambda)$ $\qquad = Z$

$$Y = \sum_{i=1}^{n} x_i \sim Ga(\alpha = n, \beta = \tfrac{1}{\lambda})$$

$$\Rightarrow Z = \frac{1}{Y} \sim \text{Invers-Gamma}(\alpha = n, \beta = \tfrac{1}{\lambda})$$

$$E[Z] = \frac{1}{\beta(\alpha - 1)} = \frac{\lambda}{n-1}$$

s. 30 i blå bok

$$E[\hat{\lambda}] = n \cdot \frac{\lambda}{n-1} = \frac{n\lambda}{n-1}$$

Opg 4)

$$f_X(x_i) = \frac{2x}{\beta} e^{-\frac{x^2}{\beta}}$$

a) $$L(\beta) = \prod_{i=1}^{n} f_X(x_i) = \left(\frac{2}{\beta}\right)^n \left(\prod_{i=1}^{n} x_i\right) \cdot e^{-\frac{1}{\beta}\sum_{i=1}^{n} x_i^2}$$

b) $$l(\beta) = \ln L(\beta) = n\ln(2) - n\ln\beta + \sum_{i=1}^{n} \ln|x_i| - \frac{1}{\beta}\sum_{i=1}^{n} x_i^2$$

c) $$\frac{dl}{d\beta} = 0 - \frac{n}{\beta} + 0 + \frac{1}{\beta^2}\sum_{i=1}^{n} x_i^2$$

$$\frac{a}{\beta} = \frac{c'b - b'c}{b^2} = \frac{0-a}{b^2}$$

d) $SME(\hat{\beta})$

$$\frac{dl}{d\beta} = 0 \Rightarrow -\frac{n}{\beta} + \frac{1}{\beta^2} S = 0 \qquad S = \sum_{i=1}^{n} x_i^2$$

$\cdot \beta^2$

$$\hat{\beta} = \frac{1}{n}\sum_{i=1}^{n} X_i^2$$

e) $f_Y(y) = ?$, $Y = X^2$

$w(y) = \sqrt{y}$, $w'(y) = \frac{1}{2\sqrt{y}}$

$$f_Y(y) = f_X(w(y)) \cdot \frac{1}{2\sqrt{y}} = \frac{2\sqrt{y}}{\beta} e^{-\frac{y}{\beta}} \cdot \frac{1}{2\sqrt{y}}$$

$$= \frac{1}{\beta} e^{-\frac{y}{\beta}}$$

f) Sum av exp = gammafordeling

g) $E[\hat{\beta}] = \frac{1}{n} E[Z]$, $Z = \sum_{i=1}^{n} X_i^2 \sim$ gamma

$= \frac{1}{n} \cdot \alpha \cdot \beta = \beta$ $\quad \alpha = n, \beta = \beta$

h) $Var[\hat{\beta}] = \frac{1}{n^2} Var[Z] = \frac{1}{n^2} \cdot \alpha \beta^2 = \frac{\beta^2}{n}$